A Comprehensive Study on the COVID-19 Pandemic and Its Multi-Sectoral Impact

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Abstract

This project presents an integrated study of the COVID-19 pandemic and its wide-ranging effects across public health, education, research, agriculture, society, and the Indian economy. It outlines the emergence and global spread of SARS-CoV-2, the major policy responses, and the challenges created by lockdowns and mobility restrictions. The study documents how education shifted online, how research activity was reshaped by the "covidization" of scientific priorities, and how clinical care, particularly for cancer patients, was disrupted. The agricultural sector, labour dynamics, social behaviour, and mental health were also deeply affected. A focused analysis of the pandemic in India recounts the distinct features of the first and second waves and the strain on healthcare systems. Finally, a mutation-frequency study using publicly available genomic data traces the evolution of SARS-CoV-2 in India, highlighting the rise of key lineages such as D614G and Delta. Together, these findings provide a consolidated view of how COVID-19 reshaped multiple sectors and altered scientific and social landscapes.

Introduction to COVID-19 Pandemic

The global COVID-19 pandemic (also known as the coronavirus pandemic), caused by Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), began with an outbreak in Wuhan, China, in December 2019. Soon afterward, it spread to other parts of Asia and then worldwide in early 2020. The World Health Organization (WHO) declared the outbreak a public health emergency of international concern (PHEIC) on 30 January 2020, and assessed it as having become a pandemic on 11 March. According to WHO, the public health emergency caused by COVID-19 ended in May 202.

COVID-19 symptoms ranged from asymptomatic to deadly, but most commonly included fever, sore throat, nocturnal cough, and fatigue. Transmission of the virus was often through airborne particles. Mutations produced many strains (variants) with varying degrees of infectivity and virulence. COVID-19 vaccines were also being developed rapidly and deployed to the general public beginning in December 2020, made available through government and international programmes such as COVAX, aiming to provide vaccine equity. Treatments included novel antiviral drugs and symptom control. Common mitigation measures during the public health emergency included travel restrictions, lockdowns, business restrictions and closures, workplace hazard controls, mask mandates, quarantines, testing systems, and contact tracing of the infected.

Impact of COVID-19 Pandemic on Education System

In the second week of March, the state governments started the closure of schools and colleges across the country, in order to control the spread of the coronavirus. The period of closure was initially for a month but was soon indefinitely extended. During this period however, various crucial competitive exams and entrance tests of universities, board examinations, semester examinations in universities, nursery school admissions as well as admission process in universities were hampered.

The closure of the schools affected the structure of learning and schooling. Firstly, it affected the teaching and assessment methodologies. A handful of private schools were able to adopt online teaching methods immediately. On the other side, low-income private and government schools had to face complete closure. In order to ensure the continuity of teaching in institutes and universities, online teaching methodology was immediately adopted. Many schools and universities shifted their traditional examination system from

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offline to online assessment tools for evaluation. A looming concern in the mind of everyone, especially freshly graduated students, was the effect of the pandemic on the rate of employment and the withdrawal of job offers from corporate.

In case of online learning in rural areas, majority of the students were engaged in assisting parents in farm activities such as agriculture, tending to cattle and household chores. Most of them did not have access to smartphones or TV at home in addition to poor Internet connectivity. There was negligible income due to closure of businesses. The data package (costs) were comparatively high against average income earned, and continuous access to the Internet was costly. This put the teachers in a dilemma as to what tools to adopt. Some thought that pre-recorded videos could help; however, that would restrict interactions. Designing a proper system to fit the learning needs and convenience of all students turned out to be a tricky affair.

Although there were overwhelming challenges for educators, schools, institutes and the government regarding online education, from a different perspective there were several opportunities created by the COVID-19 pandemic for the distant plans of implementing online learning system. It forged a stronger connection between teachers and parents than ever before. The homeschooling required parents to support the student's learning academically and economically. Children with disabilities needed additional and special support during the pandemic. The use of online platforms such as Google Classroom, Zoom, virtual learning environment and social media and various group forums like Telegram and WhatsApp were explored and tried for teaching and learning for the first time ever to continue education. Many educational organizations were offering their tools and solutions for free to help and support teaching and learning in a more interactive and engaging environment. Online learning has now provided the opportunity to teach and learn in innovative ways unlike the teaching and learning experiences in the normal classroom setting.

The Impact of COVID-19 on Research

Adaptation

In 2020, the rapid global spread of SARS–CoV-2 led research institutions worldwide to adopt emergency measures. Many universities restricted activities to essential work, and research unrelated to COVID-19 slowed markedly as COVID-focused projects expanded with unprecedented urgency. Internal funding mechanisms were introduced to accelerate pandemic-related studies. For instance, Pennsylvania State University supported 17 proposals with a total of $1.2 million. Several universities directed their efforts toward addressing shortages in personal protective equipment and responding to healthcare needs. Rutgers University established COVID-19 testing facilities capable of processing up to 10,000 samples per day. Engineers at the University of Wisconsin collaborated with local industries to manufacture face shields, and multiple groups developed rapid point-of-care testing technologies.

The Rutgers Clinical and Translational Science Award (CTSA) Hub demonstrated exceptional speed by generating a full experimental design, securing regulatory approval, and initiating enrolment of 750 volunteers within 12 days. National support further strengthened these efforts: the Coronavirus Preparedness and Response Supplemental Appropriations Act, enacted on March 6, 2020, allocated funds across federal health agencies for vaccine development, treatments, diagnostic testing, surveillance, and scientific investigations into the virus.

Research operations at Rutgers were shifted almost entirely to remote modes, maintaining strict social distancing. New projects requiring on-site presence were paused, except those directly related to COVID-19, which were strongly encouraged. These rapid institutional adaptations and discoveries played a crucial role in responding to the pandemic.

The Lost Trials

Strict lockdowns severely disrupted laboratory and clinical research. Animal studies were notably affected: institutions reduced animal numbers, restricted breeding to essential levels, and limited activities to "critical" research while prioritizing worker safety. Researchers were advised to cryopreserve rodent strains, and adjustments were made to standard operating protocols.

According to Evaluate Pharma, clinical trials were among the earliest and hardest hit sectors. In April 2020 alone, more than 160 clinical trials were suspended due to pandemic-related issues, and by July 2020 the number of suspended trials had risen to nearly 1000. As a consequence, clinical researchers faced restricted access to research infrastructure, and publications involving non-COVID clinical trials declined by approximately 20 percent. These disruptions disproportionately affected early-career scientists who relied on ongoing trials and continuous data collection.

Covidization of Research

The pandemic led to a substantial reorientation of global research priorities. There was, on average, a 6.5-fold increase in COVID-19-related Medical Subject Headings (MeSH) and a 10–12 percent decrease in publications unrelated to MeSH categories. Using PubMed data comprising 3.6 million research papers, a difference-in-differences analysis revealed a clear shift: three months after the outbreak began, the volume of COVID-19 research was already five times greater than the number of publications generated during the H1N1 influenza outbreak. Top clinical journals also exhibited a marked rise in COVID-related articles, coinciding with declines in other research areas.

Fraser and collaborators showed that COVID-related preprints received more citations than those unrelated to COVID. Hook and Porter observed that research funding increasingly favoured pandemic-related topics. Biomedical publications from PubMed (which contains over 34 million citations) were categorised using MeSH to define narrow research fields. A difference-in-differences econometric approach was applied, treating the pandemic as a natural experiment by comparing "treated" COVID-related fields with "control" non-COVID fields before and after January 2020. This allowed estimation of the pandemic's causal impact on publication volume by removing pre-existing differences between the groups.

Impact of COVID-19 on Cancer

Immunosuppression and The Ventilator Dilemma

Immunosuppression is a state where the immune system is weakened increasing a patients risk of diseases. Immunosuppressed condition of some cancer patients may expose them to serious infections followed by treatment delay and hospitalizations. Liang and others found out that cancer patients are at a higher risk of severe infections. They generally need mechanical ventilation or ICU admission. Thus Cancer patients are more susceptible to complications of COVID 19.

Justifying this fact in 2020 a retrospective cohort study recruited 28 cancer patients with confirmed COVID 19 from 3 hospitals in Wuhan ,China to assess the risk factors associated with ICU admission ,mechanical ventilation or death. This study concluded that COVID 19 infected cancer patients give poor clinical outcomes. The mortality rate was also higher among them. Cancer treatment within 2 weeks of COVID 19 diagnosis caused acute respiratory distress syndrome in 28.6% people, septic shock in 3.6 %people and acute myocardial inflation in 3.6 % people and several others .Also 28.6 % of the study patients developed COVID 19 infection while being hospitalised mainly because of nosocomial transmission. The authors of this study suggested the cancer patients should undergo thorough screening for COVID 19 infection and avoid immunosuppressive therapy in case of COVID 19 infection.

The Treatment Gap

In several clinical treatments of cancer were disrupted due to COVID invasion. Also whether to continue cancer therapy or to stop it was debatable as the risk of cancer after halt of cancer therapy is controversial. In some lethal cancers, timely diagnosis and treatment are necessary. Whereas some early stage cancers delaying treatment due to COVID can be an option. It was recommended that patients undergoing curative cancer therapy should continue their treatment despite having a potential risk of COVID 19 infection during anticancer therapy. Surgeries were delayed in some cases based on proper clinical judgement. Thus the pros and cons of treatment delay were properly assessed before suggesting treatment delay to the patients.

There was a severe deterioration of the condition of cancer patients after COVID crisis ,thus 3 essential strategies were implemented at the earliest to mitigate the crisis and to reduce the chances of future infection. Those were

- Postponing chemotherapy treatment in case of stable cases and at high-risk locations.
- Implementing strict personal precautions in cancer patients and cancer survivors.
- Providing more intensive care and treatment to COVID 19 infected cancer patients.

Along with these some other measures were also taken like online medical counselling ,treatment of critical cases, using outpatient health care facilities to treat non-small lung cancer patients whenever feasible etc. Kutikov et al proposed a guide which can be used to decide whether the cancer treatment was necessary or can be delayed.

Some examples of low risk conditions where treatment can be delayed :Low grade lymphomas, Nonmelanoma skin cancers, Chronic hematological cancers, most thyroid cancers, prostate cancers etc

The initiative that were taken at the health institutions to reduce this crisis were

- Implementation of robust infection and environment control
- Some institutions banned travel to high risk areas. Dana Farber Cancer Institute and MD Anderson Cancer institute banned all business travel to some affected areas and encouraged their employees to restrict travel.
- Some instances of virtual health services were organised to reduce exposure.
- Some health care instituted were well prepared with isolation areas.

Some examples of high risk conditions where treatment delay affect clinical outcomes : High-grade sarcoma, head and neck cancers, testicular and rectal cancers, gynecologic, rectal, and lung cancers.

- To prevent the growth of COVID 19 pandemic ,physicians have transitioned millions of visits to telehealth. This is mainly dependent on video applications, patient portals and phone calls. Both patients and oncologists have widely adapted to this new procedure.
- In some cases healthcare could not be delivered and for that treatment delay was introduced. A huge amount of oncology treatment requires systematic chemotherapy for patients with high risk cancer. Chemotherapy has limited survival benefits thus delay is suggested when

the patient is affected with COVID. For example: for most metrastatic solid tumors, chemotherapy beyond the third regimen does not increase the survival chances thus oncologists often suggest supportive care instead.

- There are some cases where delay has a moderate clinically important influence on quality of life and survival chances. Research proves that continuing a drug called rituximab for three years after a stem cell transplant helps patients improve their survival chances by 9 %. . However, because this treatment weakens the immune system, some doctors are not recommending it to prevent fatal COVID-19 infections, even though it increases the risk of the cancer returning.
- Now consider cancers for which the treatment that has potential to cure and cannot be delayed without side effects. The list of these high risk cancers was mentioned before. In spite of the presence of risks oncologists are not modifying these treatments as the cancers itself are lethal and maybe more lethal than COVID.

Impact of COVID-19 Pandemic on Agriculture

Agricultural markets were, like all other sectors, affected by the large drop in consumer spending brought on by the COVID-19 pandemic. To contain the spread of COVID-19, just like how other countries did, India imposed a complete lockdown in March which coincided with the peak of harvesting season of Rabi crops in India mainly in the north-west which posed significant losses to the farmers. Although there were relaxations to the agriculture sector during lockdown, however transport constraints, mobility restrictions and lack of labour due to reverse-migration of labour to their native places were major problems faced by the farmers. Some farmers called it a worser situation than what had occurred during the demonetization in 2016. Before the onset of the pandemic, the rural economy of India was witnessing a decline in incomes of mainly casual workers along with declining rural wages. Some rays of hope were seen in January 2020 when food prices started rising but all hopes collapsed with this pandemic. Horticulture and Foodgrains production were impacted differently. Horticulture faced the brunt more because of their perishable nature whereas food grains are non-perishable and apart from problems in harvesting and labour shortage, it was not impacted much. With declining demand and reduction in exports of fruits and vegetables, horticulture was hit hard. Similarly, floriculture was also affected because of less demands due to shut down of religious places, postponement of marriages, and so forth. In livestock (milk, meat, eggs), milk was the major contributor that was impacted and fortunately, had stability during the lockdown. Fishing and aquaculture were heavily affected as compared to horticulture and other associated activities.

COVID-19 and Its Impact on Society

Psychological effects of the pandemic and quarantine measures are evident among all groups of people. For most children, the daily school routine was a psychological comfort. Therefore, when schools were closed, mental health symptoms gradually rose. The effects were more pronounced among students that were more isolated due to social distancing and a lot of them developed severe and mild anxiety disorder symptoms. A possible solution was playing online games; however this led to sitting in front of a computer or mobile phone for long periods of time, leading to formation of unhealthy habits among children. Among students, fear of the future, especially concerning educational attainment and qualification for jobs, became abundant. Recent studies in psychological science showed that the pandemic increased mental health problems like post-traumatic stress disorder (PTSD), confusion, loneliness feeling, boredom during and after the quarantine too. Older people are battling with larger health risks as well as are more prone to be less fit for supporting themselves in isolation.

Studies show that during the pandemic, women were more vulnerable to losing their jobs as compared to men. The gender division of domestic work in India is already one of the most unequal globally, though an increase of 0.5 to 4 hours in men's proportion of housework post-lockdown has been observed. Covid-19 has also raised the income inequality between low skilled workers, and other vulnerable sections, and those with higher and advanced degrees.

A report by National Commission for Women (NCW) said that there was a 45% increase in the number of cases of domestic violence against women as compared to pre-lockdown cases. The real situation, however, might be more dangerous because many women from rural areas especially do not file a case and raise their voice and are scared of their husbands and family. The overall crime rate, however, has decreased significantly in India and globally too.

The pandemic marked the biggest 'reverse migration' India has experienced since its independence in 1947. 'Reverse migration' refers to the act/process or movement of individuals back to their local spots who prior moved to urban communities or urban regions looking for employment and to gain bread-butter for their families. Limited employment opportunities, financial and health crisis, uncertainty of future crisis and many more factors contributed to reverse migration.

Covid-19 saw an increase in work from home culture. Staying home meant more sleep, less stress and more time to spend with families. This also reflected in less pollution levels as the quantities of fuel consumption decreased due to less flights and less use of transportation.

COVID-19: Impact on the Indian Economy

The economic impact of COVID-19 on India cannot be understood as a sudden shock to an otherwise healthy economy. When the pandemic reached India in early 2020, the economy was already losing momentum. Growth had been slowing for several years, unemployment was rising, poverty reduction had stalled, and fiscal policy remained tightly constrained. COVID-19 and the subsequent lockdown did not create these problems; they exposed and intensified them.

From around 2011–12, core macroeconomic indicators such as investment, savings, exports, and bank credit showed a steady decline. This slowdown became sharper after demonetisation in 2016, which disrupted informal economic activity and employment. At the same time, the government continued to prioritise fiscal consolidation over counter-cyclical spending. As a result, when the pandemic struck, India entered the crisis with limited room to absorb a large economic shock.

Animal husbandry and poultry were hit particularly hard. Demand for milk and meat fell as restaurants and hotels shut down, and misinformation further reduced consumption. Poultry prices collapsed, forcing many producers to cull birds or shut operations altogether. These losses were not short-lived and had lasting effects on small producers.

Industrial activity contracted sharply. The Index of Industrial Production fell to historic lows in April 2020, reflecting the near-complete halt of manufacturing and construction. Micro, small, and medium enterprises were the worst affected. Surveys conducted during the lockdown revealed widespread order cancellations, cash-flow shortages, rising inventories, and a high risk of permanent shutdown for many small firms.

To understand the full scale of the damage, it is not enough to look only at sector-specific losses. Input–Output analysis shows that disruptions in one sector quickly spread to others through inter-industry links and reduced final demand. When these indirect effects are taken into account, the overall contraction in economic activity in 2020–21 appears far deeper than early official estimates suggested. Scenario-based analysis indicates that GDP growth turned sharply negative, consistent with the unprecedented collapse observed in the first quarter of the financial year.

The government responded with an economic package announced as amounting to around 10 per cent of GDP. However, most of this consisted of credit guarantees, liquidity support, and regulatory measures rather than direct fiscal spending. Actual budgetary expenditure aimed at boosting demand and employment formed only a small share of the package. In a situation marked by collapsing demand and widespread income loss, an approach centred on credit assumed that firms and households were willing and able to borrow, an assumption that was unlikely to hold.

International experience during the pandemic suggests that large-scale fiscal intervention is unavoidable in such crises. Many advanced economies relied heavily on direct income support, wage subsidies, and public spending to stabilise demand. In comparison, India's fiscal response was modest, especially given the severity of the lockdown. This raised concerns about a slow and uneven recovery, as well as long-term scarring in employment and small business viability.

To sum, the economic impact of COVID-19 on India was severe because it struck an economy already under strain. The lockdown converted a public health crisis into a full-scale economic contraction, affecting nearly every sector and disproportionately hurting vulnerable groups. The experience underscores the importance of strong counter-cyclical fiscal policy, robust social protection, and institutional resilience. Without addressing these underlying issues, temporary shocks risk turning into prolonged economic damage.

Literature Review: Tracking the Evolution of SARS-CoV-2 in India: A Mutation-Frequency Study Based on Public Genome Data

Abstract

This Literature review examines how SARS-CoV-2 evolved in India during 2020–2021 using publicly available genome sequences. After cleaning metadata and aligned FASTA files, monthly subsets of Indian sequences were analysed to identify mutations and track how often they appeared over time. The results highlight the early rise of the D614G lineage in 2020 and the later surge of Delta-associated mutations such as L452R and P681R in 2021. The work demonstrates how simple mutation-frequency tracking can reveal the major evolutionary shifts that shaped the pandemic in India.

Introduction

Before starting this topic, I went through two reports that look at SARS-CoV-2 evolution from slightly different angles. The first paper explained why certain mutations, like D614G or N501Y, kept showing up in the news, and what actually makes a variant worth worrying about. The second paper took a wider view, looking at how human coronaviruses have evolved over decades and how SARS-CoV-2 fits into this bigger story.

Both papers made one thing painfully clear: viruses don't sit still. Even a virus with proofreading enzymes, like SARS-CoV-2, finds plenty of ways to change and adapt. What caught my attention was how the authors used basic mutation-tracking to reconstruct entire evolutionary paths. Something about that felt surprisingly accessible.

So I decided to try the same, but on a smaller, student-friendly scale. I used open datasets of SARS-CoV-2 genomes from India and asked a simple question: What did the mutation landscape look like here during 2020 and 2021?

My goal wasn't to compete with major genomic surveillance groups, of course. I just wanted to see whether I could recreate the same patterns the papers discussed, like the rise of the D614G lineage in 2020 and the dominance of Delta mutations in 2021, using straightforward analysis and a bit of R code(attached in the references).

Methods

The dataset consisted of SARS-CoV-2 metadata and aligned genomic sequences downloaded from public sources. Since the metadata contained many incomplete dates, basic cleaning was needed. Missing days were replaced with mid-month placeholders to keep the timeline usable. Only sequences recorded between January 2020 and December 2021 and labelled as collected in India were kept.

The aligned FASTA file was loaded in two ways: once with Biostrings (for indexing and filtering) and once with seqinr (for position-wise mutation calling). To avoid bias from months with unusually large numbers of sequences, each month was capped at around 120 randomly sampled genomes. This produced a balanced set of roughly 1,500 sequences.

Mutations were called by comparing each genome directly against the Wuhan reference sequence. Any site where the base differed from the reference and was not marked as 'N' was counted as a mutation. Insertions and alignment gaps were ignored to avoid artefacts. Each mutation was assigned to a gene based on the standard SARS-CoV-2 coordinate map.

For each month, the number of genomes containing a given mutation was divided by the total number of sampled genomes from that month to estimate mutation frequency. The most common mutations were then visualised across time to reveal their rise or decline.

Output

1. Stacked Bar Plot: Rise of Top Mutations (2020–2021)

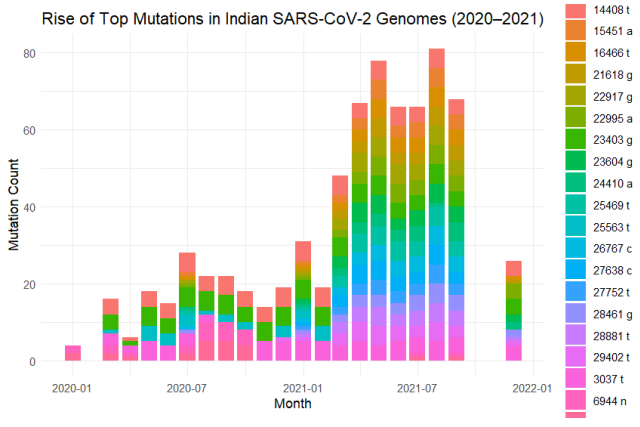

Figure 1. Rise of the most frequent SARS-CoV-2 mutations in Indian genomes (2020–2021).

Visualization:

This plot counts how many times the most common mutations appeared in each month's subsampled genomes. Each coloured block represents a different mutation, and the height of each bar reflects the total mutation load for that month.

Interpretation

Early 2020: Mutation counts are low and scattered. This reflects the fact that SARS-CoV-2 was still relatively new in India, and fewer genomic changes had accumulated.

Mid-2020: Several mutations become recurring, especially those tied to the D614G lineage (e.g., 23403G, 14408T, 3037T). This matches the global trend where D614G rapidly became dominant.

2021: Mutation counts spike sharply. This period corresponds to the rise of the Delta variant in India, which carried several defining mutations such as:

L452R (22917G)

P681R (22995A)

The tall bars around mid-2021 reflect the period where India experienced its major Delta wave, and the virus accumulated more lineage-defining changes.

2. Line Plot: Mutation Frequency Trajectories

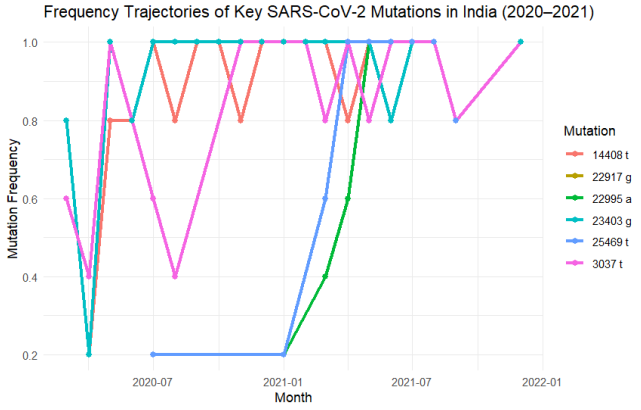

Figure 2. Temporal frequency trajectories of key SARS-CoV-2 mutations in India.

Visualization

This figure tracks how often certain key mutations appear among sampled Indian genomes each month. Instead of raw counts, this plot shows frequency , the percentage of genomes carrying each mutation.

The six mutations shown include:

23403G (D614G) — spike protein

14408T — ORF1ab

3037T — ORF1ab

22917G (L452R) — Delta lineage

22995A (P681R) — Delta lineage

25469T — ORF3a

Description

D614G, 14408T, and 3037T rise rapidly in early 2020 and stay near 100L452R and P681R remain low until early 2021, then shoot upward. 25469T fluctuates irregularly across months.

Conclusion

This project followed how SARS-CoV-2 evolved in India during 2020–2021 by tracking changes in publicly available viral genomes. Even with a simple mutation-frequency approach, the main evolutionary events of the pandemic appeared clearly in the data. The first major shift was the rapid takeover of the D614G lineage in 2020, reflected by the near-universal rise of mutations such as 23403G, 14408T and 3037T.

The second major transition arrived in 2021, when Delta-associated mutations like L452R and P681R rose sharply. Their increase matched the timing of the Delta wave in India and highlighted how quickly a fitter lineage can dominate a population.

Overall, the analysis demonstrated that even modest genomic tools can capture the broad evolutionary story of SARS-CoV-2 and how mutation patterns reflect shifting evolutionary pressures over time.

The Indian perspective of COVID-19 Outbreak

COVID-19 emerged as a viral outbreak in Wuhan, China in December 2019. The virus entered India through various international travels and then handed over to the local population. The first reported case of COVID-19 was a student returning from Wuhan, China, on 30 January 2020. She was studying in Wuhan, returned to her home in Kerala, admitted to the hospital for treatment and quarantine, discharged after proper recovery.

The COVID-19 pandemic hit India in two waves - 2020 and 2021

First Wave

On 24th March,2020 the Prime Minister announced a complete lockdown which initially planned for 14 days was extended till 30th May, 2024. A complete lockdown meant all transport services–road, air and rail–were suspended, with exceptions for transportation of essential goods, fire, police and emergency services. Educational institutions, industrial establishments and hospitality services were also suspended.

Services such as food shops, banks and ATMs, petrol pumps, other essentials and their manufacturing were exempted.

The first wave was the first of its kind with the Indian population largely unprepared to handle it. By June, 2020, the number of confirmed cases surpassed 50000 and it imposed serious pressure on the healthcare industry. The rising demand of antibiotics, prescription drugs, vaccines, ventilators, oximeters, and other medical equipment was aggravated by disruptions in the supply chain during lockdown, panic buying and stocking. The number of hospital beds could not keep up with the rising number of cases which compelled critical non-COVID patients to use homecare facilities leading to auxiliary casualties. Moreover, India had a low daily testing capacity which led to a serious underreporting of COVID-19 cases. The testing rate in India was only 0.08 tests per 1,000 people. In comparison, the United States and Italy had rates of 1.16 and 1.02 respectively. This disparity highlights the inadequacy of India's testing strategy. Frontline healthcare workers did not have access to basic protection kits like PPE, N95 masks and those who received protection were often suspecting of their quality.

As the country came to a halt, the economy was severely affected. India's GDP growth fell down 3.1% in 2020. Due to the lockdown, the manufacturing sec- tor of non-essential items in India shut down, resulting in the reduction of workforce and no possibility to hire new employees. As a result, unemployment rose from from 6.7% on 15 March to 26% on 19 April. An estimated 140 million people lost their jobs and even more had their salaries cut. Supply chains were disrupted as international trades were banned during lockdown. The brunt of this economic downfall was taken by the farmers and migrant workers who faced utmost difficulties. By May 2020, 10% of farmers could not harvest their crop in the past month and 60% of those who did harvest reported a yield loss. Major companies in India like Larsen & Toubro, Tata Motors, UltraTech Cement, etc. momentarily shut down their operations. With the factories closed, millions of daily wage labourers and migrant workers faced unemployment and had to deal with food shortages and an uncertain future. With no buses and trains, these workers were forced to walk home to their villages which led to more casualties from exhuastion and malnourishment. Later, camps were organised by the state governments to house these migrants and provide them with basic amenities. Also special trains were allocated to take them back to their homes which mitigated some of the disaster. Thus, the first wave was a new experience for all of us and we tried to resist it to the best of our abilities.

Second Wave

By October 2020, the number of daily cases had subsided and reached a steady state. Along with that, India launched its vaccination programme in January 2021 where people were being given newly developed vaccine shots two times separated by a duration of 1-4 months depending on the vaccine variant. However, things were far from normal. A new and a more contagious variant of the virus entered the country for which the country was not prepared. By early April 2021, the number of cases rose rapidly with daily cases surpassing 300000 and 2000 deaths per day.

Although various factors have been speculated to cause this havoc such as poor implementation of health and safety precautions, allowing gatherings, etc, the second wave caused a lot of problems. With a rise in cases, hospitals ran short of beds to provide to COVID patients. The number of patients requiring critical care also increased rapidly which led to shortage of necessary medicines and equipments like liquid oxygen, cryogenic tankers, etc. This led to people buying these items from illegal sources, possibly contaminated, resulting in unprecedented deaths.

Even the bodies of those passed away from the virus were not disposed off properly following safety protocols. There were repeated reports of bodies washing upon the banks of rivers which were used daily by the people. There was also a shortage of vaccines for which many people were left unvaccinated.

As the lockdowns were imposed only in select regions of high fatality, there was no significant economic impact as such.

Table 1. COVID-19 Vaccines in India

Vaccine	Developer	Dosing Schedule
Covishield	Oxford–AstraZeneca / Serum Institute of India	2 doses, 12–16 weeks apart
Covaxin	Bharat Biotech + ICMR	2 doses, 4 weeks apart

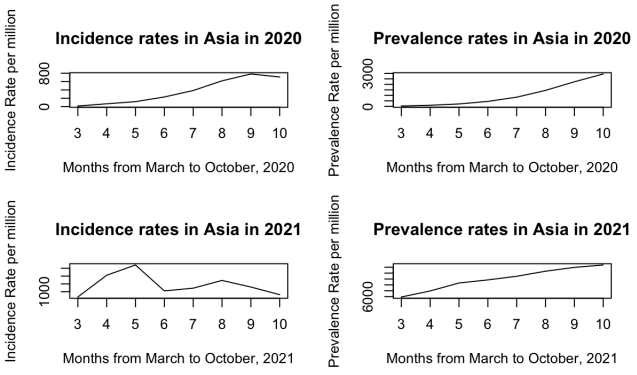

Figure 3. Incidence rate per million in Asia from March 2020 to December 2021.

Statistics All Over the World

We analysed the global progression of COVID-19 by computing incidence and prevalence rates across major continents during the periods of highest transmission. Data were obtained for the period January 2020 to December 2021, and calculations were carried out for the months March to December in both years, corresponding to intervals with substantial case loads. Above and below are the plots

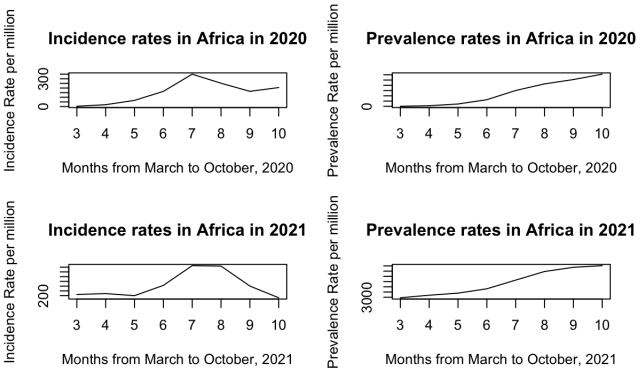

Figure 4. Incidence rate per million in Africa from March 2020 to December 2021.

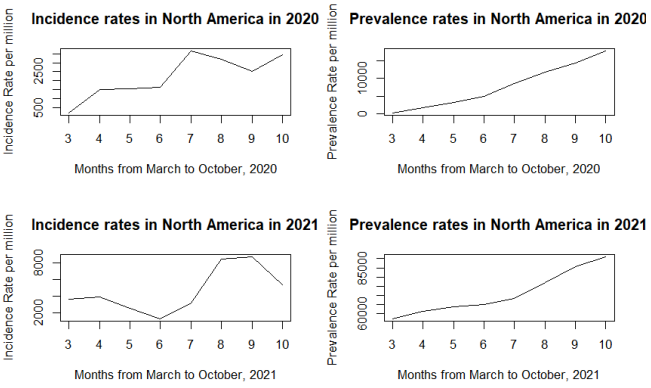

Figure 5. Incidence rate per million in North America from March 2020 to December 2021.

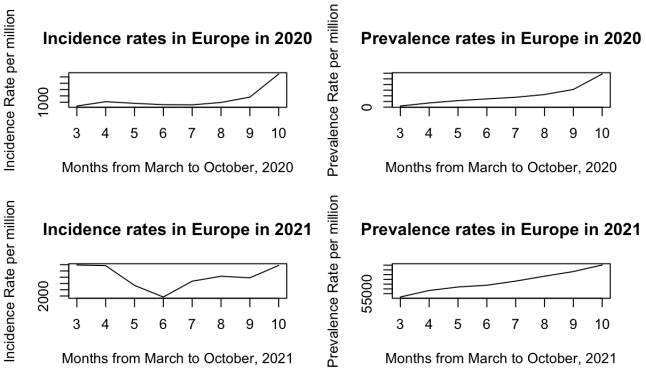

Figure 6. Incidence rate per million in Europe from March 2020 to December 2021.

representing the estimated incidence and prevalence rates across continents. Incidence increased sharply in Asia during April–May 2021, consistent with the timing of the Second Wave in India. Rates declined in most regions by October 2021 as case numbers fell. In contrast, prevalence exhibited a persistent upward trend across continents, reflecting ongoing

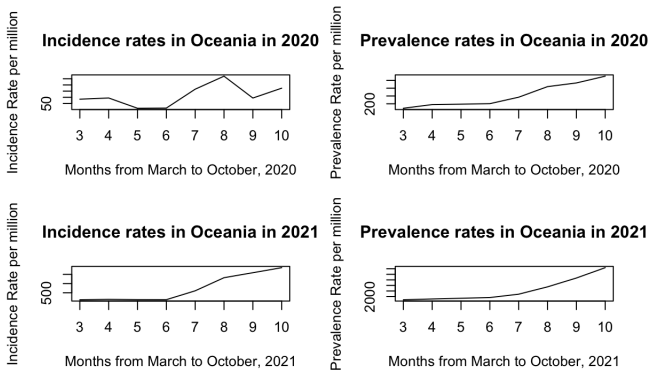

Figure 7. Incidence rate per million in Oceania from March 2020 to December 2021.

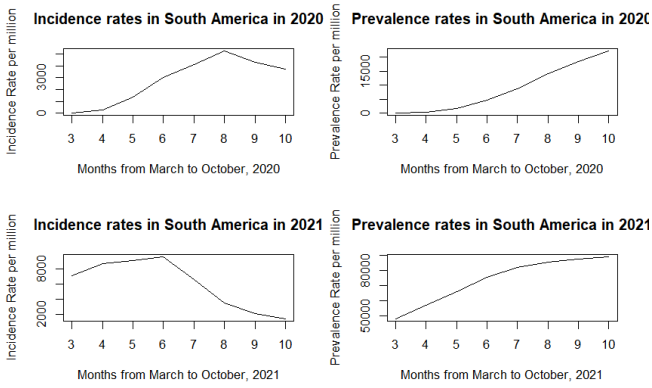

Figure 8. Incidence rate per million in South America from March 2020 to December 2021.

transmission and accumulation of active infections.

$$\text{Incidence Rate per million} = \frac{\text{Number of new cases during a specific period}}{\text{Population at risk during that period}} \times 100{,}000{,}000$$

$$\text{Prevalence Rate per million} = \frac{\text{Number of existing cases (new + old) at a given time}}{\text{Total population at that time}} \times 100{,}000{,}000$$

Authors' Note

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Author Contributions

The authors confirm contribution as follows:

- **Introduction to the COVID-19 Pandemic**: Ahan
- **Impact of the COVID-19 Pandemic on the Education System**: Ahan
- **Impact of COVID-19 on Research**: Shaswata
- **Impact of the COVID-19 Pandemic on Cancer Patients**: Shaswata
- **Impact of the COVID-19 Pandemic on Agriculture**: Sahitya
- **COVID-19 and Its Impact on Society**: Sahitya
- **COVID-19: Impact on the Indian Economy**: Shaan
- **Literature Review on the Impact of the COVID-19 Pandemic**: Shaan
- **The Indian Perspective of COVID-19 Outbreak**: Sagnik
- **Statistics All over the World**: Sagnik

All authors reviewed and approved the final manuscript.

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